

Designing for Field Awareness in Science Journalism

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Abstract

We present *News Compass*, a system to support science journalists in maintaining field awareness, i.e., their sense of what is being covered, by whom, and how. The system aggregates research papers with their news, blog, and social media coverage, along with three computational heuristics: coverage counts, information retention scores, and LLM-generated news angles. We conducted a two-week deployment study with nine practicing science journalists to evaluate their usage of the system. Our findings characterize two dimensions of field awareness journalists rely on in practice: (a) *systematic*, built through active search and scanning of material, and (b) *relational*, built through ad-hoc interactions with sources and networks. Our findings show that computational tools can support systematic awareness while raising the stakes for relational awareness, since any edge they provide can narrow once other journalists adopt the same tools. We also discuss implications for newsgathering tools in journalism, including the value of tools that surface information without interpreting it, thus leaving room for journalists' own judgment and story ideas.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools**; **Empirical studies in collaborative and social computing**.

Keywords

Computational Journalism; Newsworthiness; Awareness; Science Communication; Newsgathering

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1 Introduction and Background

The work of science journalism entails maintaining *awareness* of the *field* of news coverage, e.g., tracking what peer news outlets are covering, where gaps exist in the coverage, how coverage quality varies, where misinformation might warrant corrections, and so on. Journalists often build this awareness informally, through scanning competitor outlets, monitoring social feeds, and relying on personal

networks [14]. However, structural shifts and pressures in practice (shrinking newsrooms, faster publication cycles, and diminishing institutional support) can reduce the time and resources available to journalists [5].

Reduced time and resources incentivize practices like *pack journalism*, where stories get covered out of fear of missing what competitors cover rather than for their news value [18], as well as *churnalism*, which entails narrow coverage toward research that stakeholder institutions subsidize the coverage of (e.g., through press releases, blog posts) [8]. This state of affairs on the production side of news is further complicated by findings showing that publics increasingly receive news through social media and blogs [12], channels where key scientific concepts frequently disappear as information spreads [10]. These phenomena can run counter to science journalism's function of engaging the public with science and its multi-faceted social impacts.

In this poster, we present *News Compass*, an interactive interface designed to make the science news landscape visible and navigable for practicing journalists. The system aggregates recently published research from arXiv and OpenAlex, collects linked news, blog, and social media coverage, and presents this alongside computational heuristics such as coverage counts (after bot filtering and deduplication of syndicated stories), information retention scores [10], and LLM-generated news angles. Through these heuristics, the system aims to support journalists' field awareness of where coverage exists, where it doesn't, and where it fell short, through surfacing under-covered research and lossy coverage as opportunities for their work.

We conducted a two-week deployment study with nine practicing science journalists who incorporated the system into their regular workflows, and analyzed data from onboarding surveys, weekly check-in surveys, usage logs, and semi-structured exit interviews. Our findings characterize the *dimensions of field awareness* that drive journalists' practice—systematic versus relational access to information—and show how heightened systematic awareness can raise the value of relational awareness as a competitive edge. We also trace how journalists move *from awareness to stories*, learning that experienced journalists tended to use coverage patterns combinatorially, and that some saw potential for meta-level reporting on journalism's own collective behavior, such as identifying content farms or tracking which topics are consistently misrepresented. Finally, we examine how journalists *negotiate with computational heuristics*, e.g., calibrating trust through tool usage, appropriating features for purposes beyond their design goals, and valuing certain constraints as enabling their own judgment.

Together, these findings apply Dourish and Bellotti's [4] notion of *awareness*—developed for synchronous, collaborative settings—to an asynchronous and competitive context. Science journalists monitor each others' work in the interest of differentiating their



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own coverage, but also endeavor to build on each others' reporting to help shape public understanding of science. This reframing suggests design opportunities that we explore at the end.

2 System Design

We designed *News Compass* as a browser-based interface that aggregates recent scientific research alongside its news, blog, and social media coverage. This coverage is presented through computational heuristics designed to support journalists' field awareness (Figure 1). Journalists can customize different filters for their dashboard view, and do so either in each session, or through configuring specific filters and using their browser bookmarks to revisit them.

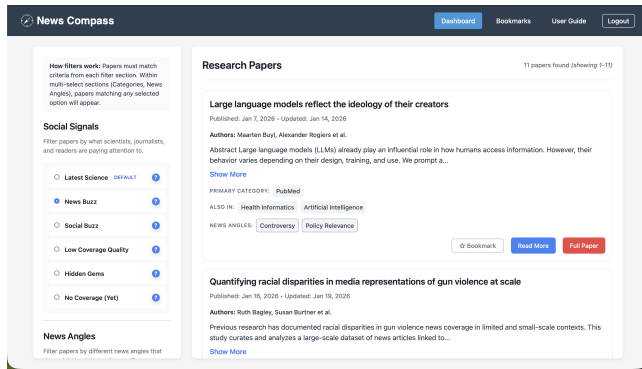


Figure 1: The *News Compass* dashboard, showing research papers with metadata, news angle badges, and the sidebar with social signal presets.

2.1 Design Goals

We formulated three design goals, which we broadly grounded in the concept of awareness as understanding the activities of others to contextualize one's own work [4], and specified based on prior work that has illustrated the competitive, time-sensitive, monitoring-heavy aspects of science journalism [7, 13, 14] and news discovery [3, 15].

- **DG1: Field Visibility.** Make the coverage landscape visible by enabling quick viewing of what research has been covered, by whom, through what channels, and with what quality.
- **DG2: Newsworthiness Evaluation.** Help journalists assess story opportunities relative to the coverage landscape, through the use of designed heuristics to surface coverage gaps, quality variation, and potential framings.
- **DG3: Configurability.** Enable journalists to navigate the landscape based on their beats, personal interests, and outlet requirements, through preset views and combinatorial filters.

2.2 Computational Heuristics

We drew on prior research to operationalize the notion of “newsworthiness” through three different computational heuristics (DG1, DG2). We motivate and describe the design of each below.

Coverage counts report the number of distinct news stories, blog posts, and social media posts (Reddit, Bluesky, LinkedIn) linked

to each paper, after filtering automated accounts and deduplicating syndicated stories. These statistics can help ascertain aspects of newsworthiness of a potential news story, such as the *shareworthiness* (i.e., would a news story on this topic attract attention on social media) [17] and *exclusivity* (i.e., has anyone else covered this new research study already?) [1]. Additionally, the ability to quickly view and navigate to social media posts around research could help journalists *discover interviewees* for their stories [13].

Information retention scores measure how well a news story preserves key concepts from the paper's abstract [10]. We extract ranked keyphrases using TextRank [11] and assess their survival in coverage text using fuzzy string matching, followed by labels that classify news stories as Low, Medium, or High quality accordingly (Figure 2). Low retention signals that existing coverage may have oversimplified or missed key findings, which journalists could treat as an *opportunity for elaboration or correction*. This score aims to generally support public-interest reporting [6].

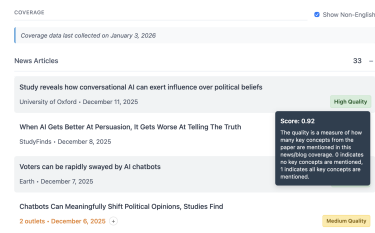


Figure 2: Coverage section of an article's page, showing news stories with information retention quality badges, and a tooltip explaining how the score is computed.

News angles are LLM-generated potential framings for each paper across different types (e.g., impact, controversy, policy relevance). These offer a way to filter and brainstorm across different newsworthiness criteria [9]. We selected GPT-5-nano for its strong calibration between its reported confidence scores (on a scale of 1 – 10, as per our instruction prompts) and human quality ratings, following an internal human evaluation study conducted by the first author ($\rho = 0.61, p < .001$). We filter its outputs to surface only angles with reported confidence scores ≥ 8 , which raises mean human-rated quality from 3.78 to 5.0 out of 7 in our human evaluation dataset, and is geared at improving the precision of the system's angle labels.

2.3 Data Collection and Display

Research papers are collected from arXiv and OpenAlex APIs via a twice-daily pipeline, and coverage metadata is collected via the Altmetric API. The *News Compass* interface offers six preset views (*Latest Science*, *News Buzz*, *Social Buzz*, *Hidden Gems*, *Low Coverage Quality*, *No Coverage Yet*), combinatorial filtering by angle, category, date, and publication status, keyword search, and bookmarking, as in Figure 3 (DG3). Papers are collected regardless of existing coverage; the *No Coverage (Yet)* preset specifically surfaces those that have not yet received any.

A key limitation here is that Altmetric surfaces coverage linked to specific papers, but does not distinguish between a story that

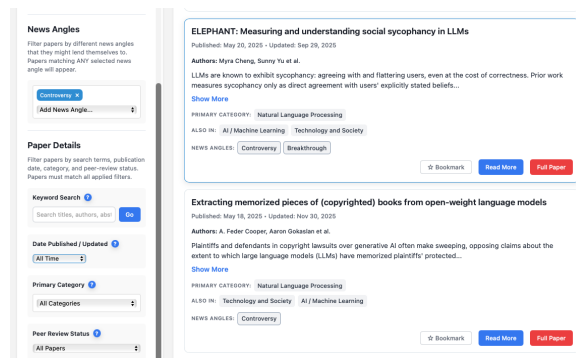


Figure 3: A list of research papers filtered by the “Controversy” news angle. The News Angles panel in the sidebar (left) supports multi-select filtering, and papers matching any selected angle are returned (OR logic).

centers a paper and one that mentions it in passing. When a news story is not solely centered on the paper Altmetric links it to, its retention score may not be accurate measure of the story’s broader information retention, across the body of work it covers. Addressing this would require moving toward heuristics organized around topics or stories rather than individual papers, which we leave to future work.

3 Study and Findings

We recruited $N = 9$ science journalists (5 freelance, 4 staff; upto 15+ years experience; beats including computing, technology, health, medicine, environmental science, and physical sciences) through professional organizations and social media outreach. Each participant used the tool for a two-week period, and we configured paper collection to include their beats for that period. We analyzed interview transcripts and open-ended survey responses via inductive thematic analysis [2], and triangulated findings with weekly Likert ratings and usage logs. The study protocol was approved by Northwestern University’s IRB (Study #STU00224608).

Through thematic analysis, we developed three broad themes, and while a full exploration of each is beyond the scope of this work, we summarize the key aspects of each theme below.

3.1 Modes of Field Awareness

Participants valued *News Compass* for *systematizing* what would otherwise require manual searches across multiple sources, but we also find that their sense of awareness of the field is built through modes the system only partially addresses. In addition to the *systematic* awareness, journalists also relied on *relational* awareness built through sources and professional networks, which surfaces newsworthy science before it is published and offers exclusivity that *News Compass* cannot match. This exclusivity is why P1 hesitated to recommend the tool to competing colleagues, since a shared edge is no edge at all. P7 articulated a potential outcome of wider adoption: as tools like *News Compass* become more widely available, they place a greater premium on relational awareness, since gathering

information before it becomes public is what can differentiate a journalists’ coverage.

3.2 From Awareness to Stories

We observed that experienced journalists used *News Compass* combinatorially, i.e., they described coming in with existing ideas and using the tool to validate, extend, or challenge them rather than generate leads from scratch. Novice journalists, on the other hand, used the tool to explore potentially newsworthy stories rather than having particular topics or themes in mind beyond the specifics of their beat. Some journalists like P2 and P5 also saw potential for the tool to enable meta-level reporting on journalism’s own collective behavior, such as through identifying content farms that republish the same low-quality stories under the guise of being local news outlets, building a thesis of how science moves through public discourse, or tracking which topics are consistently misrepresented across the coverage ecosystem.

3.3 Negotiating with Computational Heuristics

Participants used their domain expertise to test and calibrate trust rather than accepting heuristics at face value. The information retention score, in particular, invited this kind of scrutiny given the empirical claim it made: P6 raised concerns about whether keyword-matching could reward AI-generated text, and P9 noted that verbatim reproduction of press-releases would score well precisely because it retains source terminology. This critical engagement reflects deliberate use of transparency measures around the retention score, e.g., via highlighting abstract key terms, explaining the calculation. News angle labels, by contrast, attracted comparatively little pushback; participants were told during onboarding that these labels were LLM-generated, and our interpretation is that participants treated them as aids for navigating the data.

Participants also valued what *News Compass* didn’t do as much as what it did. Rather than synthesizing findings or recommending papers, the system surfaces information and leaves interpretation to the journalist. Participants noted that this restraint could preserve space for their own ideation, as P7 put it: “the more the tool itself starts prompting story ideas and things, the more it might infringe on [creativity] [...] it’s just giving me a curated set of information that then I am processing.”

4 Discussion

Our findings apply the CSCW notion of awareness [4] to an asynchronous, competitive setting, and in doing so blur the lines of Tenenberg et al.’s [16] distinction between *I-awareness* (awareness of others’ activities to inform individual action) and *we-awareness* (awareness within a collective working toward shared goals) in the context of journalism. *News Compass* was designed around *I-awareness*, but participants often reached toward *we-awareness* in the meta-level reporting use cases described above (Section 3.2). This suggests a design opportunity for field awareness that enables *individual* journalists—and potentially even editors, or civil society organizations—to work toward the *collective* goals and values of journalism, i.e., reporting in the public interest.

Journalism however is also a competitive endeavor; we find that widespread access to systematic field awareness tools can raise the

stakes for relational awareness, since the competitive edge a tool provides narrows once competitors adopt it too. This may especially hurt novice journalists, who lack the trust-based relationships that experienced journalists cultivate over years of practice. Future work might then explore the impacts of systems like *News Compass* on competition and professional development, with an eye towards impacts based on whether journalists have existing networks. There is an unresolved tension here between awareness that enables collective goals but also threatens individuals' competitiveness. We do not see an easy resolution here since exclusivity is too embedded in how newsworthiness is currently defined for any single tool to sidestep it, but we name this tension explicitly as a provocation for future work.

Limitations of the work include the small sample size, two-week window, and US-based sample. The information retention score also proved to be a narrow proxy for quality, as the findings show.

5 Conclusion

The public interest function of science journalism depends on journalists knowing what has been covered, missed, and distorted. We designed *News Compass* to enable this form of awareness, and engaged a two-week deployment study with practicing journalists. Our findings show that field awareness in journalism spans systematic and relational dimensions in tension with each other, and computational support for one can raise the stakes for the other. Rather than offer a clean resolution, our findings ask CSCW's frameworks of awareness [4, 16] to account for individual and collective goals, and systematic and relational knowledge, in conjunction.

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